



ZYROO INTERNSHIP PROGRAM

DATA ANALYTICS INTERNSHIP

WEEK 5

Advanced Business Intelligence & Decision Analytics

Project: Ride Analytics & Revenue Intelligence Platform

Project Progress

Week 2: Ride demand, locations, customer behavior, peak periods, and new vs repeat customers were analyzed.

Week 3: Revenue patterns and driver performance were analyzed using operational and financial KPIs.

Week 4: These findings were transformed into an interactive Power BI dashboard with KPIs, visuals, and filters.

Current Stage: Week 5 upgrades the dashboard into an advanced analytical and decision-support system. Week 6 will be the final end-to-end completion, validation, presentation, and evaluation.

Goal: Explain not only what happened, but why the observed patterns matter and what the business should investigate next.

1. Understand the Business Problem

- Move beyond descriptive charts and analyze relationships between demand, revenue, customers, drivers, cancellations, ratings, distance, ride type, and location.
- Every major conclusion must be supported by a measurable KPI, comparison, trend, or segment.

2. Advanced Data Model

- Review relationships, data types, duplicate IDs, missing keys, invalid dates, impossible distances, negative or zero fares, and inconsistent statuses.
- Create or improve a dedicated Date table where applicable and document the main relationships.
- Keep source data, calculated columns, measures, and presentation logic clearly separated.

3. Advanced DAX Measures

- Create Total Revenue, Completed Rides, Cancellation Rate, Completion Rate, Average Fare, Average Distance, Average Rating, and Revenue per Completed Ride.
- Add previous-period values, absolute change, percentage change, ranking, and contribution-percentage measures where supported.
- Validate important measures against SQL or Python calculations.

4. Revenue & Segment Intelligence

- Compare revenue by month, location, ride type, payment method, and driver.
- Identify high-volume/low-revenue and lower-volume/high-average-fare segments.
- Do not claim profitability unless cost data exists.

5. Customer Segmentation

- Create meaningful customer segments using ride frequency or available behavior data.
- Compare high-frequency and low-frequency customers, repeat-customer contribution, average fare, ride frequency, and preferred ride types.
- Document the evidence and business meaning of each segment.

6. Driver Performance Intelligence

- Build a driver view using rides, completed rides, cancellations, revenue, rating, and completion rate.
- Create a clearly documented ranking metric and compare drivers against relevant averages.
- Investigate unusual combinations such as high volume with high cancellation rate.

7. Cancellation & Operational Analysis

- Analyze cancellation rate by hour, weekday, location, ride type, and other available dimensions.
- Identify where cancellations concentrate and whether they overlap with high-demand periods.
- Clearly state that correlation does not prove causation.

8. Advanced Power BI Features

- Add a dedicated advanced-analysis page.
- Use appropriate features such as drill-through, report tooltips, bookmarks, decomposition tree, key influencers, conditional formatting, or dynamic titles where supported.
- Advanced does not mean overcrowded.

9. Scenario / What-If Analysis

- Create at least one practical scenario if the dataset supports it, such as estimated revenue change from a hypothetical change in completed rides, average fare, or cancellation rate.
- Clearly label scenario results as estimates and document all assumptions.

10. Business Questions

- Which periods or locations combine high demand with high cancellation rates?
- Which segments generate the largest share of completed-ride revenue?
- Which customer segments contribute most to rides or revenue?
- Which drivers differ most from the relevant benchmark?
- Which locations have strong demand but weaker revenue per completed ride?
- Which areas deserve further investigation before the final project?

11. Advanced Business Insights

- Produce at least 8 evidence-based insights.
- Include findings covering demand, revenue, customers, drivers, and cancellations.
- Use actual values and percentages from the dataset wherever possible.

12. Business Recommendations

- Produce at least 5 practical recommendations linked directly to findings.
- Recommendations may address peak-period supply, cancellations, retention, driver operations, pricing analysis, or location strategy.
- Separate dataset-supported recommendations from ideas requiring additional data or testing.

13. Week 5 Deliverable

- Advanced Power BI analytical dashboard preserving the Week 4 dashboard.
- Advanced DAX and time/segment comparisons; customer segmentation; driver intelligence; cancellation analysis; scenario analysis where applicable; 8+ insights; 5+ recommendations; methodology and data-quality notes.

14. GitHub Submission

- Push Week 5 work to the existing repository.
- Include the Power BI file where allowed, screenshots, analysis notes, and SQL/Python validation where used.
- Add a Week 5 README section covering the model, major measures, advanced features, insights, limitations, and recommendations.

15. Suggested GitHub Structure

- ride-analytics-revenue-intelligence/
 - data/
 - sql/
 - python/
 - powerbi/week-04-dashboard/
 - powerbi/week-05-advanced-analytics/
 - reports/week-05-advanced-bi/
 - screenshots/week-05/
 - README.md

16. Submission Checklist

- Advanced data model reviewed
- Date table/time analysis implemented where applicable
- Advanced DAX measures created and validated
- Revenue and segmentation completed
- Customer segmentation completed
- Driver performance intelligence completed
- Cancellation analysis completed
- Advanced Power BI features added
- Scenario analysis completed where applicable
- 8+ evidence-based insights
- 5+ recommendations
- Limitations documented
- GitHub and README updated

17. Week 5 Success Criteria

- The intern can build and explain reliable advanced DAX measures.
- The dashboard responds correctly to filters and comparisons.
- Demand, revenue, customer, driver, and operational metrics are connected.
- Findings are evidence-based and clearly separated from assumptions.

- Another analyst can understand and review the work from the documentation.

18. Final Week 5 Flow

- Week 4 Dashboard → Advanced Data Model → DAX & KPI Layer → Segmentation → Driver & Cancellation Intelligence → Scenario Analysis → Evidence-Based Insights → Business Recommendations → Week 6 Final Project

19. Next Stage: Week 6

- Week 6 will be the final and most comprehensive stage: final validation, advanced dashboard refinement, executive reporting, complete business recommendations, full GitHub documentation, final presentation, and project evaluation.

20. Official ZYROO Links

Website	https://zyroo.org
Internships	https://zyroo.org/internships
LinkedIn	https://www.linkedin.com/company/zyr0-co/
Contact	https://zyroo.org/contact